INES ALONSO

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Machine learning, agentic AI and data pipelines for scientific data · Available full time from January 2027

EDUCATION

Florida Polytechnic University, GPA 3.95/4.0 — *Lakeland, FL, USA*
Bachelor of Science: Computer Science, Minor in Applied Mathematics — **Graduating: December 2026**

University of Barcelona — *Barcelona, Spain*
Undergraduate physics coursework (three semesters); transferred to Florida Polytechnic — **Transferred 2024**

Relevant coursework: Machine Learning, Natural Language Processing, Data Structures, Algorithm Design & Analysis, Numerical Methods, Computational Linear Algebra, Quantum Information & Computing.

EXPERIENCE

Research Intern in Computational Biophysics — *Barcelona, Spain*
Institute of Photonic Sciences (ICFO), SPIE@ICFO María Yzuel Fellowship — **May 2025 – Present**

- Built an **out-of-core Python pipeline** that streams **~80 GB** of simulation data (~750,000 frames × **26,796 features**, 175 trajectories) through TICA covariance estimation with no downsampling or feature pruning.
- Trained a five-layer **VAMPnet in PyTorch** to learn nonlinear slow coordinates, and built Markov state models two independent ways (clustering-based and physics-defined states) to cross-check results.
- Designed the **validation and statistical controls**: Chapman–Kolmogorov tests, held-out cross-replica validation, a known-zero control, and false-positive calibration over ~200 randomized comparisons.

Quantum Software Research Assistant — *Lakeland, FL*
Florida Polytechnic University, Physics Department — **February 2024 – Present**

- First author and co-developer of **q-sc-SF-EOM**, a quantum algorithm for ground and excited states of molecules (Qiskit, PySCF, ADAPT-VQE); poster at IEEE Quantum Week 2026.
- Cut circuit cost at equal accuracy: the costliest matrix element drops from **52 to 16 CNOTs**; for LiH, results stay **within 0.1 eV** of the exact (FCI) result at **3.3× fewer CNOTs** than the standard method.
- Benchmarked against exact solvers (FCI, CASCI) across five molecules, and built a completeness analysis that explains when the method returns spin-contaminated states.

PROJECTS

Agentic MSM Benchmark Engine | *Independent · Python, LLM APIs, Lean 4, Pydantic, pytest* — **July 2026 – Present**

- Built a **three-agent LLM system** (Orchestrator / Optimizer / Validator) that proposes data-analysis configurations; a result is accepted only when it passes **deterministic checks against exact physics**, never on the LLM's judgment.
- Formalized the physics ground truth in **Lean 4 (Mathlib)**: 7 statements about the double-well potential, 2 proved so far (one with Axiomatic AI's **ax-prover**), with axiom checks and a Lean–Python consistency test.
- Validated the analysis engine against an exact reference: two metastable states recovered in **42/42 runs** and measured rates within **~3%** of exact across the main temperature range.
- **Audited my own system:** found the textbook formula the checker used was **6–10% off**, so it had accepted biased results; replaced it with an exact oracle, re-scored every run, and reported overconfident error bars (**2 of 42** contain the true value).

Improved Tools for Mobile-Based 3D Scene Capture, Threshold360 | *Research Lead* — **Sep 2025 – May 2026**

- Led a **distributed 3D reconstruction pipeline** (FastAPI, CUDA GPUs, Gaussian Splatting, Nerfstudio) that turns mobile captures into virtual tours in under 60 minutes.

Hierarchical Reasoning Model for Multi-hop QA | *4-person team, NLP course, PyTorch* — **November 2025**

- Designed the **token-level module** of a 60.6M-parameter Hierarchical Reasoning Model variant for HotpotQA, in which sentence-level and token-level modules exchange information through bidirectional cross-attention.

HONORS & PRESENTATIONS

SPIE@ICFO María Yzuel Fellowship (12-month research fellowship) · **NanoFlorida 2026** Lightning Round talk (selected, 20 of 80 posters) · **SHPE 2026 STEM Research Competition** (selected poster) · **IBM Qiskit Global Summer School** Quantum Excellence Badge · Florida Poly Merit Scholarship · President's List (3 terms)

LEADERSHIP

Vice President of Internal Affairs, *Society of Hispanic Professional Engineers (SHPE)* — **Jan 2025 – Jul 2026**

- Ran 10 professional-development workshops and 8 heritage events; liaison to the Provost's Office and Career Services.

Sustainability President, *Student Government Association, Florida Polytechnic* — **May 2025 – Jul 2026**

- Founded the Sustainability Agency and led 8 ambassadors; built a 100 m² campus garden on a $2,500 budget; ran events with 200+ attendees.

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, Git/GitHub, Linux, Docker, REST APIs (FastAPI), Pydantic, pytest, Lean 4 (Mathlib), LaTeX
ML & data: PyTorch, scikit-learn, NumPy/SciPy, deeptime, PyEMMA, MDTraj, CUDA, LLM APIs (multi-agent systems)
Scientific: Qiskit, PySCF, MATLAB · **Languages:** Spanish (native), Catalan (native), English (fluent), French (intermediate)